CHAPTER

FOUR

SHARED MEMORY PARALLELISM

4.1 What Is Shared?

The term **shared memory** means that the processors all share a common address space. Say this is occurring at the hardware level, and we are using Intel Pentium CPUs. Suppose processor P3 issues the instruction

```
movl 200, %ebx
```

which reads memory location 200 and places the result in the EAX register in the CPU. If processor P4 does the same, they both will be referring to the same physical memory cell. (Note, however, that each CPU has a separate register set, so each will have its own independent EAX.) In non-shared-memory machines, each processor has its own private memory, and each one will then have its own location 200, completely independent of the locations 200 at the other processors' memories.

Say a program contains a global variable **X** and a local variable **Y** on share-memory hardware (and we use shared-memory software). If for example the compiler assigns location 200 to the variable **X**, i.e. `&X = 200`, then the point is that all of the processors will have that variable in common, because any processor which issues a memory operation on location 200 will access the same physical memory cell.

On the other hand, each processor will have its own separate run-time stack. All of the stacks are in shared memory, but they will be accessed separately, since each CPU has a different value in its SP (Stack Pointer) register. Thus each processor will have its own independent copy of the local variable **Y**.

To make the meaning of "shared memory" more concrete, suppose we have a bus-based system, with all the processors and memory attached to the bus. Let us compare the above variables **X** and **Y** here. Suppose again that the compiler assigns **X** to memory location 200. Then in the machine language code for the program, every reference to **X** will be there as 200. Every time an instruction that writes to **X** is executed by a CPU, that CPU will put 200 into its Memory Address Register (MAR), from which the 200 flows out on the address lines in the bus, and goes to memory. This will happen in the same way no matter which CPU it is. Thus the same physical memory location will end up being accessed, no matter which CPU generated the reference.

By contrast, say the compiler assigns a local variable **Y** to something like ESP+8, the third item on the stack (on a 32-bit machine), 8 bytes past the word pointed to by the stack pointer, ESP. The OS will assign a different ESP value to each thread, so the stacks of the various threads will be separate. Each CPU has its own ESP register, containing the location of the stack for whatever thread that CPU is currently running. So, the value of **Y** will be different for each thread.

4.2 Memory Modules

Parallel execution of a program requires, to a large extent, parallel accessing of memory. To some degree this is handled by having a cache at each CPU, but it is also facilitated by dividing the memory into separate **modules** or **banks**. This way several memory accesses can be done simultaneously.

In this section, assume for simplicity that our machine has 32-bit words. This is still true for many GPUs, in spite of the widespread use of 64-bit general-purpose machines today, and in any case, the numbers here can easily be converted to the 64-bit case.

Note that this means that consecutive words differ in address by 4. Let's thus define the word-address of a word to be its ordinary address divided by 4. Note that this is also its address with the lowest two bits deleted.

4.2.1 Interleaving

There is a question of how to divide up the memory into banks. There are two main ways to do this:

(a) **High-order interleaving:** Here consecutive words are in the same bank (except at boundaries). For example, suppose for simplicity that our memory consists of word-addresses 0 through 1023, and that there are four banks,

M0 through M3. Then M0 would contain word-addresses 0-255, M1 would have 256-511, M2 would have 512-767, and M3 would have 768-1023.

(b) **Low-order interleaving:** Here consecutive addresses are in consecutive banks (except when we get to the right end). In the example above, if we used low-order interleaving, then word-address 0 would be in M0, 1 would be in M1, 2 would be in M2, 3 would be in M3, 4 would be back in M0, 5 in M1, and so on.

Say we have eight banks. Then under high-order interleaving, the first three bits of a word-address would be taken to be the bank number, with the remaining bits being address within bank. Under low-order interleaving, the three least significant bits would be used to determine bank number.

Low-order interleaving has often been used for **vector processors**. On such a machine, we might have both a regular add instruction, ADD, and a vector version, VADD. The latter would add two vectors together, so it would need to read two vectors from memory. If low-order interleaving is used, the elements of these vectors are spread across the various banks, so fast access is possible.

A more modern use of low-order interleaving, but with the same motivation as with the vector processors, is in GPUs (Chapter 14).

High-order interleaving might work well in matrix applications, for instance, where we can partition the matrix into blocks, and have different processors work on different blocks. In image processing applications, we can have different processors work on different parts of the image. Such partitioning almost never works perfectly—e.g. computation for one part of an image may need information from another part—but if we are careful we can get good results.

4.2.2 Bank Conflicts and Solutions

Consider an array **x** of 16 million elements, whose sum we wish to compute, say using 16 threads. Suppose we have four memory banks, with low-order interleaving.

A naive implementation of the summing code might be

```
1  parallel for thr = 0 to 15
2     localsum = 0
3     for j = 0 to 999999
4        localsum += x[thr*1000000+j]
5     grandsum += localsum  // critical section
```

In other words, thread 0 would sum the first million elements, thread 1 would sum the second million, and so on. After summing its portion of the array, a thread

would then add its sum to a grand total. (The threads *could* of course add to **grandsum** directly in each iteration of the loop, but this would cause too much traffic to memory, thus causing slowdowns.)

Suppose for simplicity that there is one address per word (it is usually one address per byte).

Suppose also for simplicity that the threads run in lockstep, so that they all attempt to access memory at once. On a multicore/multiprocessor machine, this may not occur, but it in fact typically *will* occur in a GPU setting.

A problem then arises. To make matters simple, suppose that **x** starts at an address that is a multiple of 4, thus in bank 0. (The reader should think about how to adjust this to the other three cases.) On the very first memory access, thread 0 accesses **x[0]** in bank 0, thread 1 accesses **x[1000000]**, also in bank 0, and so on—and *these will all be in memory bank 0*! Thus there will be major conflicts, hence major slowdown.

A better approach might be to have any given thread work on every sixteenth element of **x**, instead of on contiguous elements. Thread 0 would work on **x[1000000]**, **x[1000016]**, **x[10000032**,...; thread 1 would handle **x[1000001]**, **x[1000017]**, **x[10000033**,...; and so on:

```
1  parallel for thr = 0 to 15
2     localsum = 0
3     for j = 0 to 999999
4        localsum += x[16*j+thr]
5     grandsum += localsum
```

Here, consecutive threads work on consecutive elements in **x**.[1] That puts them in separate banks, thus no conflicts, hence speedy performance.

In general, avoiding bank conflicts is an art, but there are a couple of approaches we can try.

- We can rewrite our algorithm, e.g. use the second version of the above code instead of the first.
- We can add **padding** to the array. For instance in the first version of our code above, we could lengthen the array from 16 million to 16000016, placing padding in words 1000000, 2000001 and so on. We'd tweak our array indices in our code accordingly, and eliminate bank conflicts that way.

In the first approach above, the concept of **stride** often arises. It is defined to be the distance betwwen array elements in consecutive accesses by a thread. In our

[1] Here thread 0 is considered "consecutive" to thread 15, in a wraparound manner.

original code to compute **grandsum**, the stride was 1, since each array element accessed by a thread is 1 past the last access by that thread. In our second version, the stride was 16.

Strides of greater than 1 often arise in code that deals with multidimensional arrays. Say for example we have two-dimensional array with 16 columns. In C/C++, which uses row-major order, access of an entire column will have a stride of 16. Access down the main diagonal will have a stride of 17.

Suppose we have b banks, again with low-order interleaving. You should experiment a bit to see that an array access with a stride of s will access s different banks if and only if s and b are relatively prime, i.e. the greatest common divisor of s and b is 1. This can be proven with group theory.

Another strategy, useful for collections of complex objects, is to set up **structs of arrays** rather than **arrays of structs**. Say for instance we are working with data on workers, storing for each worker his name, salary and number of years with the firm. We might naturally write code like this:

```
1 struct {
2    char name[25];
3    float salary;
4    float yrs;
5 } x[100];
```

That gives a 100 structs for 100 workers. Again, this is very natural, but it may make for poor memory access patterns. Salary values for the various workers will no longer be contiguous, for instance, even though the **struct**s are contiguous. This could cause excessive cache misses.

One solution would be to add padding to each **struct**, so that the salary values are a word apart in memory. But another approach would be to replace the above arrays of **struct**s by a **struct** of arrays:

```
1 struct {
2    char name[100];
3    float salary[100];
4    float yrs[100];
5 }
```

4.2.3 Example: Code to Implement Padding

As discussed above, array padding is used to try to get better parallel access to memory banks. The code below is aimed to provide utilities to assist in this.

Details are explained in the comments.

```
1
2  // routines to initialize, read and write
3  // padded versions of a matrix of floats;
4  // the matrix is nominally mxn, but its
5  // rows will be padded on the right ends,
6  // so as to enable a stride of s down each
7  // column; it is assumed that s >= n
8
9  // allocate space for the padded matrix,
10 // initially empty
11 float *padmalloc(int m, int n, int s) {
12    return(malloc(m*s*sizeof(float)));
13 }
14
15 // store the value tostore in the matrix q,
16 // at row i, column j; m, n and
17 // s are as in padmalloc() above
18 void setter(float *q, int m, int n, int s,
19       int i, int j, float tostore) {
20    *(q + i*s+j) = tostore;
21 }
22
23 // fetch the value in the matrix q,
24 // at row i, column j; m, n and s are
25 // as in padmalloc() above
26 float getter(float *q, int m, int n, int s,
27       int i, int j) {
28    return *(q + i*s+j);
29 }
```

4.3 Interconnection Topologies

4.3.1 SMP Systems

A Symmetric Multiprocessor (SMP) system has the following structure:

- The Ps are processors, e.g. off-the-shelf chips such as Pentiums.
- The Ms are **memory modules**. These are physically separate objects, e.g. separate boards of memory chips. It is typical that there will be the same

number of Ms as Ps.

- To make sure only one P uses the bus at a time, standard bus arbitration signals and/or arbitration devices are used.
- There may also be **coherent caches**, which we will discuss later.

4.3.2 NUMA Systems

In a **Nonuniform Memory Access** (NUMA) architecture, each CPU has a memory module physically next to it, and these processor/memory (P/M) pairs are connected by some kind of network.

Each P/M/R set here is called a **processing element** (PE). Note that each PE has its own local bus, and is also connected to the global bus via R, the router.

Suppose for example that P3 needs to access location 200, and suppose that high-order interleaving is used. If location 200 is in M3, then P3's request is satisfied by the local bus.[2] On the other hand, suppose location 200 is in M8. Then the R3 will notice this, and put the request on the global bus, where it will be seen by R8, which will then copy the request to the local bus at PE8, where the request will be satisfied. (E.g. if it was a read request, then the response will go back from M8 to R8 to the global bus to R3 to P3.)

It should be obvious now where NUMA gets its name. P8 will have much faster access to M8 than P3 will to M8, if none of the buses is currently in use—and if say the global bus is currently in use, P3 will have to wait a long time to get what it wants from M8.

Today almost all high-end MIMD systems are NUMAs. One of the attractive features of NUMA is that by good programming we can exploit the nonuniformity. In matrix problems, for example, we can write our program so that, for example, P8 usually works on those rows of the matrix which are stored in M8, P3 usually works on those rows of the matrix which are stored in M3, etc. In order to do this, we need to make use of the C language's & address operator, and have some knowledge of the memory hardware structure, i.e. the interleaving.

4.3.3 NUMA Interconnect Topologies

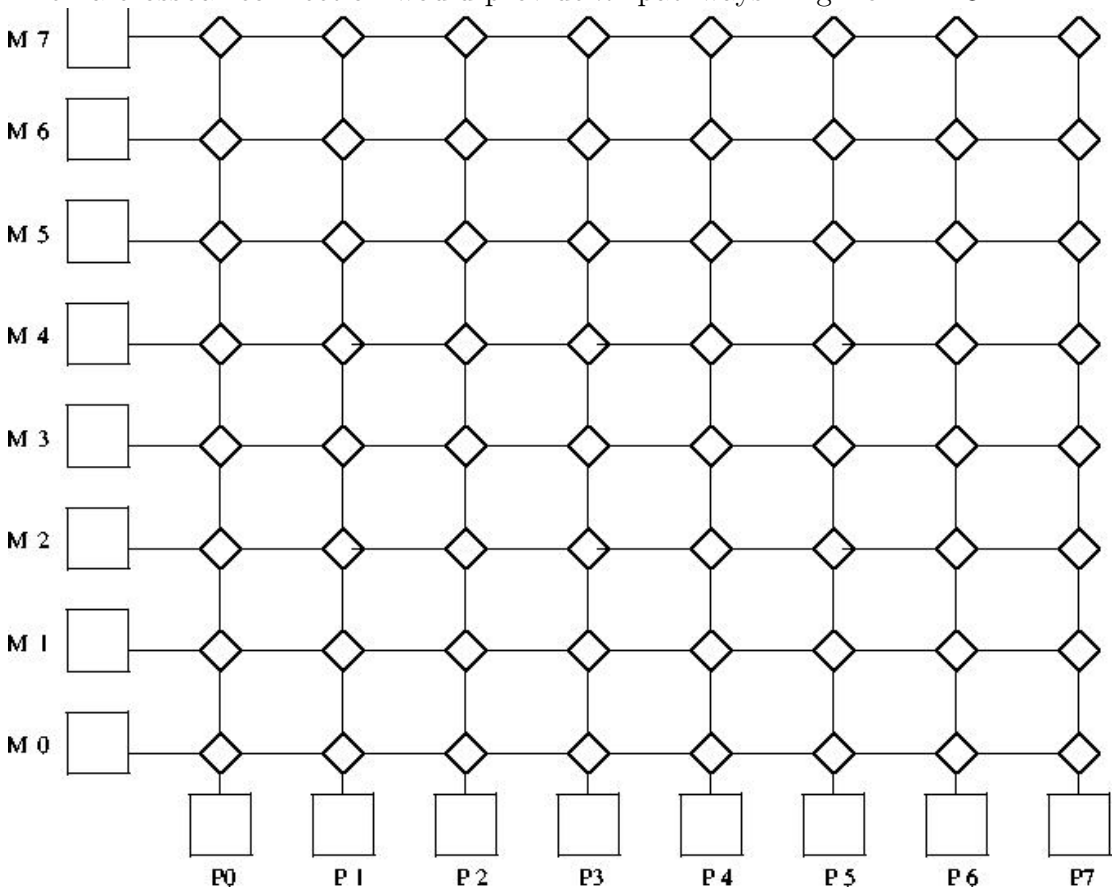
The problem with a bus connection, of course, is that there is only one pathway for communication, and thus only one processor can access memory at the same

[2] This sounds similar to the concept of a cache. However, it is very different. A cache contains a local copy of some data stored elsewhere. Here it is the data itself, not a copy, which is being stored locally.

time. If one has more than, say, two dozen processors are on the bus, the bus becomes saturated, even if traffic-reducing methods such as adding caches are used. Thus multipathway topologies are used for all but the smallest systems. In this section we look at two alternatives to a bus topology.

Crossbar Interconnects

Consider a shared-memory system with n processors and n memory modules. Then a crossbar connection would provide n^2 pathways. E.g. for n = 8:



Generally serial communication is used from node to node, with a packet containing information on both source and destination address. E.g. if P2 wants to read from M5, the source and destination will be 3-bit strings in the packet, coded as 010 and 101, respectively. The packet will also contain bits which specify which word within the module we wish to access, and bits which specify whether we wish to do a read or a write. In the latter case, additional bits are used to specify the value to be written.

Each diamond-shaped node has two inputs (bottom and right) and two outputs

(left and top), with buffers at the two inputs. If a buffer fills, there are two design options: (a) Have the node from which the input comes block at that output. (b) Have the node from which the input comes discard the packet, and retry later, possibly outputting some other packet for now. If the packets at the heads of the two buffers both need to go out the same output, the one (say) from the bottom input will be given priority.

There could also be a return network of the same type, with this one being memory → processor, to return the result of the read requests.[3]

Another version of this is also possible. It is not shown here, but the difference would be that at the bottom edge we would have the PEi and at the left edge the memory modules Mi would be replaced by lines which wrap back around to PEi, similar to the Omega network shown below.

Crossbar switches are too expensive for large-scale systems, but are useful in some small systems. The 16-CPU Sun Microsystems Enterprise 10000 system includes a 16x16 crossbar.

Omega (or Delta) Interconnects

These are multistage networks similar to crossbars, but with fewer paths.

Recall that each PE is a processor/memory pair. PE3, for instance, consists of P3 and M3.

Note the fact that at the third stage of the network (top of picture), the outputs are routed back to the PEs, each of which consists of a processor and a memory module.[4]

At each network node (the nodes are the three rows of rectangles), the output routing is done by destination bit. Let's number the stages here 0, 1 and 2, starting from the bottom stage, number the nodes within a stage 0, 1, 2 and 3 from left to right, number the PEs from 0 to 7, left to right, and number the bit positions in a destination address 0, 1 and 2, starting from the most significant bit. Then at stage i, bit i of the destination address is used to determine routing, with a 0 meaning routing out the left output, and 1 meaning the right one.

Say P2 wishes to read from M5. It sends a read-request packet, including 5 =

[3] For safety's sake, i.e. fault tolerance, even writes are typically acknowledged in multiprocessor systems.

[4] The picture may be cut off somewhat at the top and left edges. The upper-right output of the rectangle in the top row, leftmost position should connect to the dashed line which leads down to the second PE from the left. Similarly, the upper-left output of that same rectangle is a dashed lined, possibly invisible in your picture, leading down to the leftmost PE.

101 as its destination address, to the switch in stage 0, node 1. Since the first bit of 101 is 1, that means that this switch will route the packet out its right-hand output, sending it to the switch in stage 1, node 3. The latter switch will look at the next bit in 101, a 0, and thus route the packet out its left output, to the switch in stage 2, node 2. Finally, that switch will look at the last bit, a 1, and output out its right-hand output, sending it to PE5, as desired. M5 will process the read request, and send a packet back to PE2, along the same

Again, if two packets at a node want to go out the same output, one must get priority (let's say it is the one from the left input).

Here is how the more general case of N = 2^n PEs works. Again number the rows of switches, and switches within a row, as above. So, S_{ij} will denote the switch in the i-th row from the bottom and j-th column from the left (starting our numbering with 0 in both cases). Row i will have a total of N input ports I_{ik} and N output ports O_{ik}, where k = 0 corresponds to the leftmost of the N in each case. Then if row i is not the last row ($i < n-1$), O_{ik} will be connected to I_{jm}, where j = i+1 and

$$m = (2k + \lfloor (2k)/N \rfloor) \ mod \ N \tag{4.1}$$

If row i is the last row, then O_{ik} will be connected to, PE k.

4.3.4 Comparative Analysis

In the world of parallel architectures, a key criterion for a proposed feature is **scalability**, meaning how well the feature performs as we go to larger and larger systems. Let n be the system size, either the number of processors and memory modules, or the number of PEs. Then we are interested in how fast the latency, bandwidth and cost grow with n:

criterion	bus	Omega	crossbar
latency	O(1)	O($\log_2$ n)	O(n)
bandwidth	O(1)	O(n)	O(n)
cost	O(1)	O(n $\log_2$ n)	O(n^2)

Let us see where these expressions come from, beginning with a bus: No matter how large n is, the time to get from, say, a processor to a memory module will be the same, thus O(1). Similarly, no matter how large n is, only one communication can occur at a time, thus again O(1).[5]

[5]Note that the '1' in "O(1)" does not refer to the fact that only one communication can occur at a time. If we had, for example, a two-bus system, the bandwidth would still be O(1),

Again, we are interested only in "O( )" measures, because we are only interested in growth rates as the system size n grows. For instance, if the system size doubles, the cost of a crossbar will quadruple; the $O(n^2)$ cost measure tells us this, with any multiplicative constant being irrelevant.

For Omega networks, it is clear that $log_2 n$ network rows are needed, hence the latency value given. Also, each row will have n/2 switches, so the number of network nodes will be O(n $log_2 n$). This figure then gives the cost (in terms of switches, the main expense here). It also gives the bandwidth, since the maximum number of simultaneous transmissions will occur when all switches are sending at once.

Similar considerations hold for the crossbar case.

The crossbar's big advantage is that it is guaranteed that n packets can be sent simultaneously, providing they are to distinct destinations.

That is not true for Omega-networks. If for example, PE0 wants to send to PE3, and at the same time PE4 wishes to sent to PE2, the two packets will clash at the leftmost node of stage 1, where the packet from PE0 will get priority.

On the other hand, a crossbar is very expensive, and thus is dismissed out of hand in most modern systems. Note, though, that an equally troublesom aspect of crossbars is their high latency value; this is a big drawback when the system is not heavily loaded.

The bottom line is that Omega-networks amount to a compromise between buses and crossbars, and for this reason have become popular.

4.3.5 Why Have Memory in Modules?

In the shared-memory case, the Ms collectively form the entire shared address space, but with the addresses being assigned to the Ms in one of two ways:

- (a)

  High-order interleaving. Here consecutive addresses are in the same M (except at boundaries). For example, suppose for simplicity that our memory consists of addresses 0 through 1023, and that there are four Ms. Then M0 would contain addresses 0-255, M1 would have 256-511, M2 would have 512-767, and M3 would have 768-1023.

- (b)

---

since multiplicative constants do not matter. What O(1) means, again, is that as n grows, the bandwidth stays at a multiple of 1, i.e. stays constant.

Low-order interleaving. Here consecutive addresses are in consecutive M's (except when we get to the right end). In the example above, if we used low-order interleaving, then address 0 would be in M0, 1 would be in M1, 2 would be in M2, 3 would be in M3, 4 would be back in M0, 5 in M1, and so on.

The idea is to have several modules busy at once, say in conjunction with a **split-transaction bus**. Here, after a processor makes a memory request, it relinquishes the bus, allowing others to use it while the memory does the requested work. Without splitting the memory into modules, this wouldn't achieve parallelism. The bus does need extra lines to identify which processor made the request.

4.4 Synchronization Hardware

Avoidance of race conditions, e.g. implementation of locks, plays such a crucial role in shared-memory parallel processing that hardware assistance is a virtual necessity. Recall, for instance, that critical sections can effectively serialize a parallel program. Thus efficient implementation is crucial.

4.4.1 Test-and-Set Instructions

Consider a bus-based system. In addition to whatever memory read and memory write instructions the processor included, there would also be a TAS instruction.[6] This instruction would control a TAS pin on the processor chip, and the pin in turn would be connected to a TAS line on the bus.

Applied to a location L in memory and a register R, say, TAS does the following:

```
copy L to R
if R is 0 then write 1 to L
```

And most importantly, these operations are done in an **atomic** manner; no bus transactions by other processors may occur between the two steps.

The TAS operation is applied to variables used as locks. Let's say that 1 means locked and 0 unlocked. Then the guarding of a critical section C by a lock variable L, using a register R, would be done by having the following code in the program being run:

```
TRY:  TAS R,L
      JNZ TRY
  C:  ...   ; start of critical section
      ...
```

[6] This discussion is for a mythical machine, but any real system works in this manner.

```
...   ; end of critical section
MOV 0,L  ; unlock
```

where of course JNZ is a jump-if-nonzero instruction, and we are assuming that the copying from the Memory Data Register to R results in the processor N and Z flags (condition codes) being affected.

LOCK Prefix on Intel Processors

On Pentium machines, the LOCK prefix can be used to get atomicity for certain instructions: ADD, ADC, AND, BTC, BTR, BTS, CMPXCHG, DEC, INC, NEG, NOT, OR, SBB, SUB, XOR, XADD. The bus will be locked for the duration of the execution of the instruction, thus setting up atomic operations. There is a special LOCK line in the control bus for this purpose. (Locking thus only applies to these instructions in forms in which there is an operand in memory.) By the way, XCHG asserts this LOCK# bus signal even if the LOCK prefix is not specified.

For instance, say we are maintaining some global count, **total**. Instead of

```
pthread_mutex_lock(&totlock);
total ++;
pthread_mutex_unlock(&totlock);
```

we could do

```
lock inc total
```

without software locks!

Here is how we could implement a lock if needed. The lock would be in a variable named, say, **lockvar**.

```
    movl $lockvar, %ebx  # copy the address of lockvar to EBX
    movl $1, %ecx
top:
    movl $0, %eax
    lock cmpxchg (%ebx), %ecx
    jz top  # else leave the loop and enter the critical section
```

The operation CMPXCHG ("compare and exchange") has EAX as an unnamed operand. The instruction basically does (here *source* is ECX, *destination* is **lockvar**, and *c()* means "contents of")

```
if c(EAX) != c(destination)  # sorry, lock is already locked
    c(EAX) <- c(destination)
```

```
   ZF <- 0  # the Zero Flag in the EFLAGS register
else
   c(destination) <- c(source)  # lock the lock
   ZF <- 1
```

The LOCK prefix locks the bus for the entire duration of the instruction. Note that the ADD instruction involves two memory transactions. Consider

```
lock addl $3, x
```

which adds 3 to the memory location named **x**. There is one bus transaction needed to read the old value of **x**, and a second one to write the new, incremented value back to **x**. So, we are locking for a rather long time, potentially compromising performance when other threads want to access memory, but the benefits can be huge compared to using software locks.

Locks with More Complex Interconnects

In crossbar or Ω-network systems, some 2-bit field in the packet must be devoted to transaction type, say 00 for Read, 01 for Write and 10 for TAS. In a sytem with 16 CPUs and 16 memory modules, say, the packet might consist of 4 bits for the CPU number, 4 bits for the memory module number, 2 bits for the transaction type, and 32 bits for the data (for a write, this is the data to be written, while for a read, it would be the requested value, on the trip back from the memory to the CPU).

But note that the atomicity here is best done at the memory, i.e. some hardware should be added at the memory so that TAS can be done; otherwise, an entire processor-to-memory path (e.g. the bus in a bus-based system) would have to be locked up for a fairly long time, obstructing even the packets which go to other memory modules.

4.4.2 May Not Need the Latest

Note carefully that in many settings it may not be crucial to get the most up-to-date value of a variable. For example, a program may have a data structure showing work to be done. Some processors occasionally add work to the queue, and others take work from the queue. Suppose the queue is currently empty, and a processor adds a task to the queue, just as another processor is checking the queue for work. As will be seen later, it is possible that even though the first processor has written to the queue, the new value won't be visible to other processors for some time. But the point is that if the second processor does not see work in the

queue (even though the first processor has put it there), the program will still work correctly, albeit with some performance loss.

4.4.3 Fetch-and-Add Instructions

Another form of interprocessor synchronization is a **fetch-and-add** (FA) instruction. The idea of FA is as follows. For the sake of simplicity, consider code like

```
LOCK(K);
Y = ++X
UNLOCK(K);
```

Suppose our architecture's instruction set included an F&A instruction. It would add 1 to the specified location in memory, and return the old value (to **Y**) that had been in that location before being incremented. And all this would be an atomic operation.

And even better, we could allow other increments than 1.

We would then replace the code above by a library call, say,

```
FETCH_AND_ADD(X,1);
```

The C code above would compile to, say,

```
F&A X,R,1
```

where **R** is the register into which the old (pre-incrementing) value of **X** would be returned.

There would be hardware adders placed at each memory module. That means that the whole operation could be done in one round trip to memory. Without F&A, we would need two round trips to memory just for the

```
X++;
```

operation (we would load **X** into a register in the CPU, increment the register, and then write it back to **X** in memory), and then the LOCK() and UNLOCK() would need trips to memory too. This could be a huge time savings, especially for long-latency interconnects.

But the key is that this is all done in an atomic manner.

4.5 Cache Issues

If you need a review of cache memories or don't have background in that area at all, read Section **??** in the appendix of this book before continuing.

4.5.1 Cache Coherency

Consider, for example, a bus-based system. Relying purely on TAS for interprocessor synchronization would be unthinkable: As each processor contending for a lock variable spins in the loop shown above, it is adding tremendously to bus traffic.

An answer is to have caches at each processor.[7] These will store copies of the values of lock variables. (Of course, non-lock variables are stored too. However, the discussion here will focus on effects on lock variables.) The point is this: Why keep looking at a lock variable L again and again, using up the bus bandwidth? L may not change value for a while, so why not keep a copy in the cache, avoiding use of the bus?

The answer of course is that eventually L will change value, and this causes some delicate problems. Say for example that processor P5 wishes to enter a critical section guarded by L, and that processor P2 is already in there. During the time P2 is in the critical section, P5 will spin around, always getting the same value for L (1) from C5, P5's cache. When P2 leaves the critical section, P2 will set L to 0—and now C5's copy of L will be incorrect. This is the **cache coherency problem**, inconsistency between caches.

A number of solutions have been devised for this problem. For bus-based systems, **snoopy** protocols of various kinds are used, with the word "snoopy" referring to the fact that all the caches monitor ("snoop on") the bus, watching for transactions made by other caches.

The most common protocols are the **invalidate** and **update** types. This relation between these two is somewhat analogous to the relation between **write-back** and **write-through** protocols for caches in uniprocessor systems:

- Under an invalidate protocol, when a processor writes to a variable in a cache, it first (i.e. before actually doing the write) tells each other cache to mark as invalid its cache line (if any) which contains a copy of the variable.[8]

[7] The reader may wish to review the basics of caches. See for example `http://heather.cs.ucdavis.edu/~matloff/50/PLN/CompOrganization.pdf`.

[8] We will follow commonly-used terminology here, distinguishing between a *cache line* and a *memory block*. Memory is divided in blocks, some of which have copies in the cache. The cells

Those caches will be updated only later, the next time their processors need to access this cache line.

- For an update protocol, the processor which writes to the variable tells all other caches to immediately update their cache lines containing copies of that variable with the new value.

Let's look at an outline of how one implementation (many variations exist) of an invalidate protocol would operate:

In the scenario outlined above, when P2 leaves the critical section, it will write the new value 0 to L. Under the invalidate protocol, P2 will post an invalidation message on the bus. All the other caches will notice, as they have been monitoring the bus. They then mark their cached copies of the line containing L as invalid.

Now, the next time P5 executes the TAS instruction—which will be very soon, since it is in the loop shown above—P5 will find that the copy of L in C5 is invalid. It will respond to this cache miss by going to the bus, and requesting P2 to supply the "real" (and valid) copy of the line containing L.

But there's more. Suppose that all this time P6 had also been executing the loop shown above, along with P5. Then P5 and P6 may have to contend with each other. Say P6 manages to grab possession of the bus first.[9] P6 then executes the TAS again, which finds L = 0 and changes L back to 1. P6 then relinquishes the bus, and enters the critical section. Note that in changing L to 1, P6 also sends an invalidate signal to all the other caches. So, when P5 tries its execution of the TAS again, it will have to ask P6 to send a valid copy of the block. P6 does so, but L will be 1, so P5 must resume executing the loop. P5 will then continue to use its valid local copy of L each time it does the TAS, until P6 leaves the critical section, writes 0 to L, and causes another cache miss at P5, etc.

At first the update approach seems obviously superior, and actually, if our shared, cacheable[10] variables were only lock variables, this might be true.

But consider a shared, cacheable vector. Suppose the vector fits into one block, and that we write to each vector element sequentially. Under an update policy, we would have to send a new message on the bus/network for each component, while under an invalidate policy, only one message (for the first component) would be needed. If during this time the other processors do not need to access this vector,

---

in the cache are called *cache lines*. So, at any given time, a given cache line is either empty or contains a copy (valid or not) of some memory block.

[9] Again, remember that ordinary bus arbitration methods would be used.

[10] Many modern processors, including Pentium and MIPS, allow the programmer to mark some blocks as being noncacheable.

all those update messages, and the bus/network bandwidth they use, would be wasted.

Or suppose for example we have code like

```
Sum += X[I];
```

in the middle of a **for** loop. Under an update protocol, we would have to write the value of Sum back many times, even though the other processors may only be interested in the final value when the loop ends. (This would be true, for instance, if the code above were part of a critical section.)

On the other hand, if **Sum** is the *only* word in the block that we access here, when the cache block is finally sent to other caches, the entire block must be sent under the invalidate, say 512 bytes. If we perform the above summation operation fewer than 512/8 times, the update protocol will do less global memory writing, thus tying up the bus less.

Thus the invalidate protocol works well for some kinds of code, while update works better for others. The CPU designers must try to anticipate which protocol will work well across a broad mix of applications.[11]

Now, how is cache coherency handled in non-bus shared-memory systems, say crossbars? Here the problem is more complex. Think back to the bus case for a minute: The very feature which was the biggest negative feature of bus systems—the fact that there was only one path between components made bandwidth very limited—is a very positive feature in terms of cache coherency, because it makes broadcast very easy: Since everyone is attached to that single pathway, sending a message to all of them costs no more than sending it to just one—we get the others for free. That's no longer the case for multipath systems. In such systems, extra copies of the message must be created for each path, adding to overall traffic.

A solution is to send messages only to "interested parties." In **directory-based** protocols, a list is kept of all caches which currently have valid copies of all blocks. In one common implementation, for example, while P2 is in the critical section above, it would be the **owner** of the block containing L. (Whoever is the latest node to write to L would be considered its current owner.) It would maintain a directory of all caches having valid copies of that block, say C5 and C6 in our story here. As soon as P2 wrote to L, it would then send either invalidate or update packets (depending on which type was being used) to C5 and C6 (and not to other caches which didn't have valid copies).

There would also be a directory at the memory, listing the current owners of all

[11] Some protocols change between the two modes dynamically.

blocks. Say for example P0 now wishes to "join the club," i.e. tries to access L, but does not have a copy of that block in its cache C0. C0 will thus not be listed in the directory for this block. So, now when it tries to access L and it will get a cache miss. P0 must now consult the **home** of L, say P14. The home might be determined by L's location in main memory according to high-order interleaving; it is the place where the main-memory version of L resides. A table at P14 will inform P0 that P2 is the current owner of that block. P0 will then send a message to P2 to add C0 to the list of caches having valid copies of that block. Similarly, a cache might "resign" from the club, due to that cache line being replaced, e.g. in a LRU setting, when some other cache miss occurs.

4.5.2 Example: the MESI Cache Coherency Protocol

Many types of cache coherency protocols have been proposed and used, some of them quite complex. A relatively simple one for snoopy bus systems which is widely used is MESI, which for example is the protocol used in the Pentium series.

MESI is an invalidate protocol for bus-based systems. Its name stands for the four states a given cache line can be in for a given CPU:

- Modified
- Exclusive
- Shared
- Invalid

Note that *each memory block* has such a state at *each cache*. For instance, block 88 may be in state S at P5's and P12's caches but in state I at P1's cache.

Here is a summary of the meanings of the states:

state	meaning
M	written to more than once; no other copy valid
E	valid; no other cache copy valid; memory copy valid
S	valid; at least one other cache copy valid
I	invalid (block either not in the cache or present but incorrect)

Following is a summary of MESI state changes.[12] When reading it, keep in mind again that there is a separate state for each cache/memory block combination.

[12] See *Pentium Processor System Architecture*, by D. Anderson and T. Shanley, Addison-Wesley, 1995. We have simplified the presentation here, by eliminating certain programmable options.

In addition to the terms **read hit**, **read miss**, **write hit**, **write miss**, which you are already familiar with, there are also **read snoop** and **write snoop**. These refer to the case in which our CPU observes on the bus a block request by another CPU that has attempted a read or write action but encountered a miss in its own cache; if our cache has a valid copy of that block, we must provide it to the requesting CPU (and in some cases to memory).

So, here are various events and their corresponding state changes:

If our CPU does a read:

present state	event	new state
M	read hit	M
E	read hit	E
S	read hit	S
I	read miss; no valid cache copy at any other CPU	E
I	read miss; at least one valid cache copy in some other CPU	S

If our CPU does a memory write:

present state	event	new state
M	write hit; do not put invalidate signal on bus; do not update memory	M
E	same as M above	M
S	write hit; put invalidate signal on bus; update memory	E
I	write miss; update memory but do nothing else	I

If our CPU does a read or write snoop:

present state	event
M	read snoop; write line back to memory, picked up by other CPU
M	write snoop; write line back to memory, signal other CPU now OK to do its write
E	read snoop; put shared signal on bus; no memory action
E	write snoop; no memory action
S	read snoop
S	write snoop
I	any snoop

Note that a write miss does NOT result in the associated block being brought in from memory.

Example: Suppose a given memory block has state M at processor A but has state I at processor B, and B attempts to write to the block. B will see that its copy of the block is invalid, so it notifies the other CPUs via the bus that it intends to do this write. CPU A sees this announcement, tells B to wait, writes its own copy of

the block back to memory, and then tells B to go ahead with its write. The latter action means that A's copy of the block is not correct anymore, so the block now has state I at A. B's action does not cause loading of that block from memory to its cache, so the block still has state I at B.

4.5.3 The Problem of "False Sharing"

Consider the C declaration

```
int W,Z;
```

Since **W** and **Z** are declared adjacently, most compilers will assign them contiguous memory addresses. Thus, unless one of them is at a memory block boundary, when they are cached they will be stored in the same cache line. Suppose the program writes to **Z**, and our system uses an invalidate protocol. Then **W** will be considered invalid at the other processors, even though its values at those processors' caches are correct. This is the **false sharing** problem, alluding to the fact that the two variables are sharing a cache line even though they are not related.

This can have very adverse impacts on performance. If for instance our variable **W** is now written to, then **Z** will suffer unfairly, as its copy in the cache will be considered invalid even though it is perfectly valid. This can lead to a "ping-pong" effect, in which alternate writing to two variables leads to a cyclic pattern of coherency transactions.

One possible solution is to add padding, e.g. declaring **W** and **Z** like this:

```
int W,U[1000],Z;
```

to separate **W** and **Z** so that they won't be in the same cache block. Of course, we must take block size into account, and check whether the compiler really has placed the two variables are in widely separated locations. To do this, we could for instance run the code

```
printf("%x %x\n,&W,&Z);
```

4.6 Memory-Access Consistency Policies

Though the word *consistency* in the title of this section may seem to simply be a synonym for *coherency* from the last section, and though there actually is some relation, the issues here are quite different. In this case, it is a timing issue: After one processor changes the value of a shared variable, when will that value be visible to the other processors?

There are various reasons why this is an issue. For example, many processors, especially in multiprocessor systems, have **write buffers**, which save up writes for some time before actually sending them to memory. (For the time being, let's suppose there are no caches.) The goal is to reduce memory access costs. Sending data to memory in groups is generally faster than sending one at a time, as the overhead of, for instance, acquiring the bus is amortized over many accesses. Reads following a write may proceed, without waiting for the write to get to memory, except for reads to the same address. So in a multiprocessor system in which the processors use write buffers, there will often be some delay before a write actually shows up in memory.

That's a hardware example. On the software level, compilers typically will store a program variable x in a register rather than in the memory location the compiler has chosen for that variable. Register access is fast, so this makes sense, but the compiler must produce code so that eventually the value of x is written to memory, so that other processors can use it.

The designer of a multiprocessor system must adopt some **consistency model** regarding situations like this. The above discussion shows that the programmer must be made aware of the model, or risk getting incorrect results. Note also that different consistency models will give different levels of performance. The "weaker" consistency models make for faster machines but require the programmer to do more work.

The strongest consistency model is Sequential Consistency. It essentially requires that memory operations done by one processor are observed by the other processors to occur in the same order as executed on the first processor. Enforcement of this requirement makes a system slow, and it has been replaced on most systems by weaker models.

One such model is **release consistency**. Here the processors' instruction sets include instructions ACQUIRE and RELEASE. Execution of an ACQUIRE instruction at one processor involves telling all other processors to flush their write buffers. However, the ACQUIRE won't execute until pending RELEASEs are done. Execution of a RELEASE basically means that you are saying, "I'm done writing for the moment, and wish to allow other processors to see what I've written." An ACQUIRE waits for all pending RELEASEs to complete before it executes.[13]

A related model is **scope consistency**. Say a variable, say **Sum**, is written to within a critical section guarded by LOCK and UNLOCK instructions. Then

[13]There are many variants of all of this, especially in the software distibuted shared memory realm, to be discussed later.

under scope consistency any changes made by one processor to **Sum** within this critical section would then be visible to another processor when the latter next enters this critical section. The point is that memory update is postponed until it is actually needed. Also, a barrier operation (again, executed at the hardware level) forces all pending memory writes to complete.

All modern processors include instructions which implement consistency operations. For example, Sun Microsystems' SPARC has a MEMBAR instruction. If used with a STORE operand, then all pending writes at this processor will be sent to memory. If used with the LOAD operand, all writes will be made visible to this processor.

Now, how does cache coherency fit into all this? There are many different setups, but for example let's consider a design in which there is a write buffer between each processor and its cache. As the processor does more and more writes, the processor saves them up in the write buffer. Eventually, some programmer-induced event, e.g. a MEMBAR instruction,[14] will cause the buffer to be flushed. Then the writes will be sent to "memory"—actually meaning that they go to the cache, and then possibly to memory.

The point is that (in this type of setup) before that flush of the write buffer occurs, the cache coherency system is quite unaware of these writes. Thus the cache coherency operations, e.g. the various actions in the MESI protocol, won't occur until the flush happens.

To make this notion concrete, again consider the example with **Sum** above, and assume release or scope consistency. The CPU currently executing that code (say CPU 5) writes to **Sum**, which is a memory operation—it affects the cache and thus eventually the main memory—but that operation will be invisible to the cache coherency protocol for now, as it will only be reflected in this processor's write buffer. But when the unlock is finally done (or a barrier is reached), the write buffer is flushed and the writes are sent to this CPU's cache. That then triggers the cache coherency operation (depending on the state). The point is that the cache coherency operation would occur only now, not before.

What about reads? Suppose another processor, say CPU 8, does a read of **Sum**, and that page is marked invalid at that processor. A cache coherency operation will then occur. Again, it will depend on the type of coherency policy and the current state, but in typical systems this would result in **Sum**'s cache block being shipped to CPU 8 from whichever processor the cache coherency system thinks has a valid copy of the block. That processor may or may not be CPU 5, but

[14] We call this "programmer-induced," since the programmer will include some special operation in her C/C++ code which will be translated to MEMBAR.

even if it is, that block won't show the recent change made by CPU 5 to **Sum**.

The analysis above assumed that there is a write buffer between each processor and its cache. There would be a similar analysis if there were a write buffer between each cache and memory.

Note once again the performance issues. Instructions such as ACQUIRE or MEMBAR will use a substantial amount of interprocessor communication bandwidth. A consistency model must be chosen carefully by the system designer, and the programmer must keep the communication costs in mind in developing the software.

The recent Pentium models use Sequential Consistency, with any write done by a processor being immediately sent to its cache as well.

4.7 Fetch-and-Add Combining within Interconnects

In addition to read and write operations being specifiable in a network packet, an F&A operation could be specified as well (a 2-bit field in the packet would code which operation was desired). Again, there would be adders included at the memory modules, i.e. the addition would be done at the memory end, not at the processors. When the F&A packet arrived at a memory module, our variable **X** would have 1 added to it, while the old value would be sent back in the return packet (and put into R).

Another possibility for speedup occurs if our system uses a multistage interconnection network such as a crossbar. In that situation, we can design some intelligence into the network nodes to do **packet combining**: Say more than one CPU is executing an F&A operation at about the same time for the same variable **X**. Then more than one of the corresponding packets may arrive at the same network node at about the same time. If each one requested an incrementing of **X** by 1, the node can replace the two packets by one, with an increment of 2. Of course, this is a delicate operation, and we must make sure that different CPUs get different return values, etc.

4.8 Multicore Chips

A recent trend has been to put several CPUs on one chip, termed a **multicore** chip. As of March 2008, dual-core chips are common in personal computers, and quad-core machines are within reach of the budgets of many people. Just as the

invention of the integrated circuit revolutionized the computer industry by making computers affordable for the average person, multicore chips will undoubtedly revolutionize the world of parallel programming.

A typical dual-core setup might have the two CPUs sharing a common L2 cache, with each CPU having its own L1 cache. The chip may interface to the bus or interconnect network of via an L3 cache.

Multicore is extremely important these days. However, they are just SMPs, for the most part, and thus should not be treated differently.

4.9 Optimal Number of Threads

A common question involves the best number of threads to run in a shared-memory setting. Clearly there is no general magic answer, but here are some considerations:[15]

- If your application does a lot of I/O, CPUs or cores may stay idle while waiting for I/O events. It thus makes sense to have many threads, so that computation threads can run when the I/O threads are tied up.
- In a purely computational application, one generally should not have more threads than cores. However, a program with a lot of virtual memory page faults may benefit from setting up extra threads, as page replacement involves (disk) I/O.
- Applications in which there is heavy interthread communication, say due to having a lot of lock variable, access, may benefit from setting up fewer threads than the number of cores.
- Many Intel processors include hardware for *hypertheading*. These are not full threads in the sense of having separate cores, but rather involve a limited amount of resource duplication within a core. The performance gain from this is typically quite modest. In any case, be aware of it; some software systems count these as threads, and assume for instance that there are 8 cores when the machine is actually just quad core.
- With GPUs (Chapter 14), most memory accesses have long latency and thus are I/O-like. Typically one needs very large numbers of threads for good performance.

[15] As with many aspects of parallel programming, a good basic knowledge of operating systems is key. See the reference on page **??**.

4.10 Processor Affinity

With a timesharing OS, a given thread may run on different cores during different timeslices. If so, the cache for a given core may need a lot of refreshing, each time a new thread runs on that core. To avoid this slowdown, one might designate a preferred core for each thread, in the hope of reusing cache contents. Setting this up is dependent on the chip and the OS. OpenMP 3.1 has some facility for this.

4.11 Illusion of Shared-Memory through Software

4.11.1 Software Distributed Shared Memory

There are also various shared-memory software packages that run on message-passing hardware such as NOWs, called **software distributed shared memory** (SDSM) systems. Since the platforms do not have any physically shared memory, the shared-memory view which the programmer has is just an illusion. But that illusion is very useful, since the shared-memory paradigm is believed to be the easier one to program in. Thus SDSM allows us to have "the best of both worlds"—the convenience of the shared-memory world view with the inexpensive cost of some of the message-passing hardware systems, particularly networks of workstations (NOWs).

SDSM itself is divided into two main approaches, the **page-based** and **object-based** varieties. The page-based approach is generally considered clearer and easier to program in, and provides the programmer the "look and feel" of shared-memory programming better than does the object-based type.[16] We will discuss only the page-based approach here. The most popular SDSM system today is the page-based Treadmarks (Rice University). Another excellent page-based system is JIAJIA (Academy of Sciences, China).

To illustrate how page-based SDSMs work, consider the line of JIAJIA code

```
Prime = (int *) jia_alloc(N*sizeof(int));
```

The function **jia_alloc()** is part of the JIAJIA library, **libjia.a**, which is linked to one's application program during compilation.

At first this looks a little like a call to the standard **malloc()** function, setting up an array **Prime** of size **N**. In fact, it does indeed allocate some memory. Note that each node in our JIAJIA group is executing this statement, so each node allocates

[16] The term *object-based* is not related to the term *object-oriented programming*.

some memory at that node. Behind the scenes, not visible to the programmer, each node will then have its own copy of **Prime**.

However, JIAJIA sets things up so that when one node later accesses this memory, for instance in the statement

```
Prime[I] = 1;
```

this action will eventually trigger a network transaction (not visible to the programmer) to the other JIAJIA nodes.[17] This transaction will then update the copies of **Prime** at the other nodes.[18]

How is all of this accomplished? It turns out that it relies on a clever usage of the nodes' virtual memory (VM) systems. To understand this, you need a basic knowledge of how VM systems work. If you lack this, or need review, read Section **??** in the appendix of this book before continuing.

Here is how VM is exploited to develop SDSMs on Unix systems. The SDSM will call a system function such as **mprotect()**. This allows the SDSM to deliberately mark a page as nonresident (even if the page *is* resident). Basically, anytime the SDSM knows that a node's local copy of a variable is invalid, it will mark the page containing that variable as nonresident. Then, the next time the program at this node tries to access that variable, a page fault will occur.

As mentioned in the review above, normally a page fault causes a jump to the OS. However, technically any page fault in Unix is handled as a signal, specifically SIGSEGV. Recall that Unix allows the programmer to write his/her own signal handler for any signal type. In this case, that means that the programmer—meaning the people who developed JIAJIA or any other page-based SDSM—writes his/her own page fault handler, which will do the necessary network transactions to obtain the latest valid value for **X**.

Note that although SDSMs are able to create an illusion of almost all aspects of shared memory, it really is not possible to create the illusion of shared pointer variables. For example on shared memory hardware we might have a variable like **P**:

```
int Y,*P;
...
...
P = &Y;
...
```

[17] There are a number of important issues involved with this word *eventually*, as we will see later.

[18] The update may not occur immediately. More on this later.

There is no simple way to have a variable like **P** in an SDSM. This is because a pointer is an address, and each node in an SDSM has its own memory separate address space. The problem is that even though the underlying SDSM system will keep the various copies of **Y** at the different nodes consistent with each other, **Y** will be at a potentially different address on each node.

All SDSM systems must deal with a software analog of the cache coherency problem. Whenever one node modifies the value of a shared variable, that node must notify the other nodes that a change has been made. The designer of the system must choose between update or invalidate protocols, just as in the hardware case.[19] Recall that in non-bus-based shared-memory multiprocessors, one needs to maintain a directory which indicates at which processor a valid copy of a shared variable exists. Again, SDSMs must take an approach similar to this.

Similarly, each SDSM system must decide between sequential consistency, release consistency etc. More on this later.

Note that in the NOW context the internode communication at the SDSM level is typically done by TCP/IP network actions. Treadmarks uses UDP, which is faster than TCP. but still part of the slow TCP/IP protocol suite. TCP/IP was simply not designed for this kind of work. Accordingly, there have been many efforts to use more efficient network hardware and software. The most popular of these is the Virtual Interface Architecture (VIA).

Not only are coherency actions more expensive in the NOW SDSM case than in the shared-memory hardware case due to network slowness, there is also expense due to granularity. In the hardware case we are dealing with cache blocks, with a typical size being 512 bytes. In the SDSM case, we are dealing with pages, with a typical size being 4096 bytes. The overhead for a cache coherency transaction can thus be large.

4.12 Barrier Implementation

Recall that a **barrier** is program code[20] which has a processor do a wait-loop action until all processors have reached that point in the program.[21]

[19] Note, though, that we are not actually dealing with a cache here. Each node in the SDSM system will have a cache, of course, but a node's cache simply stores parts of that node's set of pages. The coherency across nodes is across pages, not caches. We must insure that a change made to a given page is eventually propropagated to pages on other nodes which correspond to this one.

[20] Some hardware barriers have been proposed.

[21] I use the word *processor* here, but it could be just a thread on the one hand, or on the other hand a processing element in a message-passing context.

A function **Barrier()** is often supplied as a library function, e.g. **pthread_barrier_wait()**. Here we will see how to implement such a library function in a correct and efficient manner. Note that since a barrier is a serialization point for the program, efficiency is crucial to performance.

Implementing a barrier in a fully correct manner is actually a bit tricky. We'll see here what can go wrong, and how to make sure it doesn't.

In this section, we will approach things from a shared-memory point of view (which is the main place barriers are used). But the methods apply in the obvious way to message-passing systems as well, as will be discused later.

4.12.1 A Use-Once Version

```
1  struct BarrStruct {
2     int NNodes, // number of threads participating in the barrier
3         Count, // number of threads that have hit the barrier so far
4         pthread_mutex_t Lock = PTHREAD_MUTEX_INITIALIZER;
5  } ;
6
7  Barrier(struct BarrStruct *PB)
8  {  pthread_mutex_lock(&PB->Lock);
9     PB->Count++;
10    pthread_mutex_unlock(&PB->Lock);
11    while (PB->Count < PB->NNodes) ;
12 }
```

This is very simple, actually overly so. This implementation will work once, so if a program using it doesn't make two calls to **Barrier()** it would be fine. But not otherwise. If, say, there is a call to **Barrier()** in a loop, we'd be in trouble.

What is the problem? Clearly, something must be done to reset **Count** to 0 at the end of the call, but doing this safely is not so easy, as seen in the next section.

4.12.2 An Attempt to Write a Reusable Version

Consider the following attempt at fixing the code for **Barrier()**:

```
1  Barrier(struct BarrStruct *PB)
2  {  int OldCount;
3     pthread_mutex_lock(&PB->Lock);
4     OldCount = PB->Count++;
5     pthread_mutex_unlock(&PB->Lock);
6     if (OldCount == PB->NNodes-1)  PB->Count = 0;
7     else while (PB->Count < PB->NNodes) ;
8  }
```

Unfortunately, this doesn't work either. To see why, consider a loop with a barrier call at the end:

```
struct BarrStruct B;  // global variable
........
while (.......)  {
   .........
   Barrier(&B);
   .........
}
```

At the end of the first iteration of the loop, all the processors will wait at the barrier until everyone catches up. After this happens, one processor, say 12, will reset **B.Count** to 0, as desired. But if we are unlucky, some other processor, say processor 3, will then race ahead, perform the second iteration of the loop in an extremely short period of time, and then reach the barrier and increment the **Count** variable before processor 12 resets it to 0. This would result in disaster, since processor 3's increment would be canceled, leaving us one short when we try to finish the barrier the second time.

Another disaster scenario which might occur is that one processor might reset **B.Count** to 0 before another processor had a chance to notice that **B.Count** had reached **B.NNodes**.

4.12.3 A Correct Version

One way to avoid this would be to have *two* **Count** variables, and have the processors alternate using one then the other. In the scenario described above, processor 3 would increment the *other* **Count** variable, and thus would not conflict with processor 12's resetting. Here is a safe barrier function based on this idea:

```
struct BarrStruct {
   int NNodes, // number of threads participating in the barrier
       Count[2], // number of threads that have hit the barrier so far
       pthread_mutex_t Lock = PTHREAD_MUTEX_INITIALIZER;
} ;

Barrier(struct BarrStruct *PB)
{  int Par,OldCount;
   Par = PB->EvenOdd;
   pthread_mutex_lock(&PB->Lock);
   OldCount = PB->Count[Par]++;
   if (OldCount == PB->NNodes-1)  {
      PB->Count[Par] = 0;
      PB->EvenOdd = 1 - Par;
      pthread_mutex_unlock(&PB->Lock);
   }
   else {
      pthread_mutex_unlock(&PB->Lock);
      while (PB->Count[Par] > 0) ;
   }
}
```

4.12.4 Refinements

Use of Wait Operations

The code

```
else while (PB->Count[Par] > 0) ;
```

is harming performance, since it has the processor spining around doing no useful work. In the Pthreads context, we can use a condition variable:

```
1  struct BarrStruct {
2     int NNodes, // number of threads participating in the barrier
3         Count[2], // number of threads that have hit the barrier so far
4         pthread_mutex_t Lock = PTHREAD_MUTEX_INITIALIZER;
5         pthread_cond_t CV = PTHREAD_COND_INITIALIZER;
6  } ;
7
8  Barrier(struct BarrStruct *PB)
9  {  int Par,I;
10    Par = PB->EvenOdd;
11    pthread_mutex_lock(&PB->Lock);
12    PB->Count[Par]++;
13    if (PB->Count < PB->NNodes)
14       pthread_cond_wait(&PB->CV,&PB->Lock);
15    else  {
16       PB->Count[Par] = 0;
17       PB->EvenOdd = 1 - Par;
18       for (I = 0; I < PB->NNodes-1; I++)
19          pthread_cond_signal(&PB->CV);
20    }
21    pthread_mutex_unlock(&PB->Lock);
22 }
```

Here, if a thread finds that not everyone has reached the barrier yet, it still waits for the rest, but does so passively, via the wait for the condition variable **CV**. This way the thread is not wasting valuable time on that processor, which can run other useful work.

Note that the call to **pthread_cond_wait()** requires use of the lock. Your code must lock the lock before making the call. The call itself immediately unlocks that lock after it registers the wait with the threads manager. But the call blocks until awakened when another thread calls **pthread_cond_signal()** or **pthread_cond_broadcast()**.

It is required that your code lock the lock before calling **pthread_cond_signal()**, and that it unlock the lock after the call.

By using **pthread_cond_wait()** and placing the unlock operation later in the code, as seen above, we actually could get by with just a single **Count** variable, as before.

Even better, the **for** loop could be replaced by a single call

```
pthread_cond_broadcast(&PB->CV);
```

This still wakes up the waiting threads one by one, but in a much more efficient way, and it makes for clearer code.

Parallelizing the Barrier Operation

Tree Barriers

It is clear from the code above that barriers can be costly to performance, since they rely so heavily on critical sections, i.e. serial parts of a program. Thus in many settings it is worthwhile to parallelize not only the general computation, but also the barrier operations themselves.

Consider for instance a barrier in which 16 threads are participating. We could speed things up by breaking this barrier down into two sub-barriers, with eight threads each. We would then set up three barrier operations: one of the first group of eight threads, another for the other group of eight threads, and a third consisting of a "competition" between the two groups. The variable **NNodes** above would have the value 8 for the first two barriers, and would be equal to 2 for the third barrier.

Here thread 0 could be the representative for the first group, with thread 4 representing the second group. After both groups's barriers were hit by all of their members, threads 0 and 4 would participated in the third barrier.

Note that then the notification phase would the be done in reverse: When the third barrier was complete, threads 0 and 4 would notify the members of their groups.

This would parallelize things somewhat, as critical-section operations could be executing simultaneously for the first two barriers. There would still be quite a bit of serial action, though, so we may wish to do further splitting, by partitioning each group of four threads into two subroups of two threads each.

In general, for n threads (with n, say, equal to a power of 2) we would have a tree structure, with $log_2 n$ levels in the tree. The i^{th} level (starting with the root as level 0) with consist of 2^i parallel barriers, each one representing $n/2^i$ threads.

Butterfly Barriers

Another method basically consists of each node "shaking hands" with every other node. In the shared-memory case, handshaking could be done by having a global

array **ReachedBarrier**. When thread 3 and thread 7 shake hands, for instance, would reach the barrier, thread 3 would set **ReachedBarrier[3]** to 1, and would then wait for **ReachedBarrier[7]** to become 1. The wait, as before, could either be a **while** loop or a call to **pthread_cond_wait()**. Thread 7 would do the opposite.

If we have n nodes, again with n being a power of 2, then the barrier process would consist of $log_2 n$ phases, which we'll call phase 0, phase 1, etc. Then the process works as follows.

For any node i, let i(k) be the number obtained by inverting bit k in the binary representation of i, with bit 0 being the least significant bit. Then in the k^{th} phase, node i would shake hands with node i(k).

For example, say n = 8. In phase 0, node $5 = 101_2$, say, would shake hands with node $4 = 100_2$.

Actually, a butterfly exchange amounts to a number of simultaneously tree operations.